

Light at night co-distributes with incident breast but not lung cancer in the female population of Israel.

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Recent studies of shift-working women have reported that excessive exposure to light at night (LAN) may be a risk factor for breast cancer. However, no studies have yet attempted to examine the co-distribution of LAN and breast cancer incidence on a population level with the goal to assess the coherence of these earlier findings with population trends. Coherence is one of Hill's "criteria" (actually, viewpoints) for an inference of causality. Nighttime satellite images were used to estimate LAN levels in 147 communities in Israel. Multiple regression analysis was performed to investigate the association between LAN and breast cancer incidence rates and, as a test of the specificity of our method, lung cancer incidence rates in women across localities under the prediction of a link with breast cancer but not lung cancer. After adjusting for several variables available on a population level, such as ethnic makeup, birth rate, population density, and local income level, a strong positive association between LAN intensity and breast cancer rate was revealed ($p < 0.05$), and this association strengthened ($p < 0.01$) when only statistically significant factors were filtered out by stepwise regression analysis. Concurrently, no association was found between LAN intensity and lung cancer rate. These results provide coherence of the previously reported case-control and cohort studies with the co-distribution of LAN and breast cancer on a population basis. The analysis yielded an estimated 73% higher breast cancer incidence in the highest LAN exposed communities compared to the lowest LAN exposed communities.